

Cover Letter

Original submission to Physical Communication (Elsevier)

Date: August 24, 2026

To: The Editor-in-Chief, *Physical Communication* (Elsevier)

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Manuscript title: Open-Set Single-Channel Blind Source Separation: Per-Source Modulation Detection is SNR-Dependent

Manuscript type: Original Research Paper

Dear Editor,

We are pleased to submit the above manuscript for consideration as an original research article in *Physical Communication*. The paper addresses a gap that sits precisely at the physical layer: every existing single-channel blind source separation (SC-BSS) system for co-frequency communication signals assumes that the modulation format of every source is known a priori, while a practical spectrum-monitoring or cognitive-radio receiver has no such guarantee. We reframe SC-BSS as an *open-set recognition problem* and show that the standard way of measuring open-set performance—pooling detection scores across operating conditions—is actively misleading in this domain, because per-source detectability *flips with the SNR*. A training-free SNR-routed scoring rule then exploits this structure.

Key strengths of the manuscript.

1. **First open-set formulation of SC-BSS.** We formalise joint separation and per-source modulation identification with rejection of unseen modulations, and provide a benchmark: four known modulations (BPSK/QPSK/8PSK/16QAM), four held-out modulations (64QAM, $\pi/4$ -DQPSK, MSK, OFDM-QPSK), and three evaluation protocols (both sources known / one unknown / both unknown). To our knowledge no prior work has evaluated a learned separator under unseen-modulation conditions.
2. **A finding with methodological consequences.** Six standard post-hoc OOD scorers (Energy, MSP, ODIN, Prototype, VOS, Mahalanobis) are all at chance when pooled across SNR—yet strongly complementary when conditioned on SNR: geometry-based scores reach 0.84 AUROC at 10 dB and collapse below 0 dB, where logit-based scores peak at 0.74. Pooled open-set metrics can therefore be qualitatively wrong for physical-layer signals; condition-aware evaluation is required. This caution transfers to other OOD settings with a measurable operating-condition variable.
3. **A simple, principled, deployable fix.** The SNR-routed ensemble (rule fixed a priori from the validation SNR profile, not tuned on test) lifts the weighted-average AUROC from 0.526 (best single scorer) to 0.625, within 0.06 of the post-hoc oracle, at zero training cost and negligible inference overhead ($\sim 8.6\text{K}$ extra parameters on a $\sim 69\text{K}$ -parameter backbone). It degrades gracefully under SNR-estimation error (≥ 0.576 at $\sigma = 6\text{ dB}$) and is stable across a $100\times$ range of carrier-frequency separations (5 Hz–500 Hz).
4. **Evidence quality.** Every reported number is multi-seed (five seeds for all main results, three for ablations); six post-hoc OOD scorers from two families are compared; two training-side ablations (center loss, embedding dimension) show the effect lives in the backbone features; and a held-out reference-pool check confirms that the reported AUROCs

change by less than 0.003 when the OOD reference statistics are estimated on data disjoint from the test protocols.

5. **Reproducibility.** The training, evaluation, and figure-generation code and the synthetic data generator are publicly available at https://github.com/BinNong/single_channel_source_seperation/tree/main/paper3_open_set; no external datasets are required.

Headline numbers (five seeds). On the open-set benchmark with a lightweight complex-valued backbone ($\sim 69\text{K}$ parameters): pooled OOD AUROC is ≈ 0.50 for *all six* scorers (the metric trap); per-SNR breakdown reveals the complementary structure; the SNR-routed ensemble reaches weighted-average AUROC 0.625 ± 0.031 vs. 0.526 ± 0.043 (best single scorer) and a post-hoc oracle of 0.681 ± 0.038 ; closed-set separation remains intact (SI-SDR turns positive at $\text{SNR} \geq 10\text{ dB}$), confirming that the difficulty lies in detection, not separation.

Related submissions. A companion manuscript on *closed-set* separation architecture design (the complex-valued CNN with complex squeeze-and-excitation used as the backbone here) is currently under review at *Wireless Personal Communications* (Springer). The two manuscripts share no results and address orthogonal problems: the companion designs the separator; the present manuscript formulates and analyses open-set detection on top of it, and is fully self-contained.

Originality and ethics.

- The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal.
- All authors have approved the submission; the authors have no conflicts of interest to declare.
- No human subjects or animal experiments are involved; all experiments use synthetic signals generated by our data generator.

Author contributions. B. Nong conceived the open-set formulation, ran the multi-seed experiments and analyses, and drafted the manuscript. W. Fu contributed to the conceptualisation of the detection framework. Z. Jiang contributed to the experimental investigation. All authors reviewed and approved the final manuscript.

Funding. This work received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Suggested reviewers. We have no specific reviewer suggestions; we trust the editorial board to identify expert reviewers in physical-layer communications and machine-learning-based signal processing.

Thank you for considering our manuscript. We look forward to hearing from you.

Sincerely,

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